# Multimodal Fake News & Image Tampering Diagnostic Report

Generated on 2026-09-22 10:09:09 IST | Project Forensic Audit
Analysis performed at 2026-09-22 10:06:51 IST

## 1. Executive Summary & Verdict

| | |
|---|---|
| Final Classification Verdict | **FAKE** |
| Overall Multimodal Risk Score | 0.7362 (0.0=Real, 1.0=Fake) |
| Prediction Confidence | 81.81% |
| Processing Time | 7.75 seconds |

This submission was classified as likely fabricated or misleading, with high confidence (81.8%). The article's text contributed most to this verdict (53% of the fused score), primarily because no attribution to an official source, study, or named authority was found in the text. Additional supporting evidence: text model attribution flagged these tokens as most fake-indicative: claim.

## 2. Evidence Summary

• No attribution to an official source, study, or named authority was found in the text.
• Text model attribution flagged these tokens as most fake-indicative: claim.
• Error Level Analysis found no strong splicing/re-compression signal (anomaly score 0.068).
• Image classifier assigned this image an elevated manipulation-risk score.
• Noise-inconsistency analysis found unusually uneven noise levels across the image, a pattern sometimes associated with compositing.
• Edge-density analysis found an uneven distribution across image quadrants, consistent with a possible splice boundary.
• Simulated propagation graph shows an elevated proportion of suspicious/bot-like spreader accounts (29%).
• Graph Attention Network flagged the propagation structure as high-risk.

## 3. Model Results & Fusion Weights

| Modality | Model Engine | Risk Score | Weight | Relative Impact |
|---|---|---|---|---|
| Text Module | TF-IDF / BERT Transformer | 0.9668 | 0.40 | 52.5% |
| Image Module | ResNet50 / ELA / pHash | 0.5341 | 0.35 | 25.4% |
| Graph Module | PyG Graph Attention (GAT) | 0.6503 | 0.25 | 22.1% |

## 4. Analyzed Article Content

*"many researchers gave hypothesis that pyramids in egypt is built by aliens some people also claim that they saw ufo building it"*

## 5. Text Explanation — Top Contributing Tokens

SHAP/coefficient attribution identified the following tokens as most influential in the text-model verdict: **claim, researchers**.

## 6. Image Explanation

Error Level Analysis showed no strong splicing signal (anomaly score 0.0679). Perceptual hash: `c44438381c8142a6`. A GradCAM attention heatmap localizing the regions most influential to the ResNet-50 verdict was generated for this image (see dashboard for the visual overlay).

Forensic observations: noise levels vary unusually across regions of the image (coefficient of variation 1.00), a pattern sometimes associated with compositing content from a different source; edge density is unevenly distributed across the image quadrants (imbalance ratio 4.3x), which can indicate a localized splice boundary.

## 7. Propagation Analysis Summary

Simulated propagation graph: 45 nodes, 135 cascade edges, 6 detected communities, diffusion depth 4, 29% suspicious/bot-like accounts. Graph Attention Network propagation risk score: 0.6503.

## 8. Fusion Analysis

The final verdict is a weighted linear combination of the unimodal risk scores (see Section 3), calibrated via Platt scaling fit on the project's labeled demo corpus. Any modality not submitted for this analysis defaults to a neutral 0.5 score rather than being fabricated. The modality with the largest 'Relative Impact' in the table above contributed most to the final classification.

## 9. Final Decision & Recommendations

This content shows characteristics consistent with misinformation. Before sharing or acting on it: verify the claim against at least one established fact-checking organization or primary source, check whether the accompanying image appears elsewhere in a different context (reverse image search), and treat unattributed claims and emotionally-charged language as a caution signal.